WASP-52b

R-0451

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_171012` · created 2026-07-19T17:30:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458038.6523 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.013
Transit depth (Rp/Rs)^2	2.8 +/- 0.43 [%]
Orbital Inclination (inc)	85.21 +/- 0.8
Airmass coefficient 1 (a1)	1.0544 +/- 0.0088
Airmass coefficient 2 (a2)	-0.044 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.83 %
Best Comparison Star	#3 - [301, 370]
Optimal Aperture	2.42
Optimal Annulus	9.3
Transit Duration (day)	0.0748 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.013 vs 0.1646 (deviation 0.18σ)
Duration fit vs published	0.0748 d vs 0.0758 d (deviation 0.21σ)
Mid-time O–C	1.9 min
Depth SNR	12.8
Residual scatter	0.83 %

Light curve

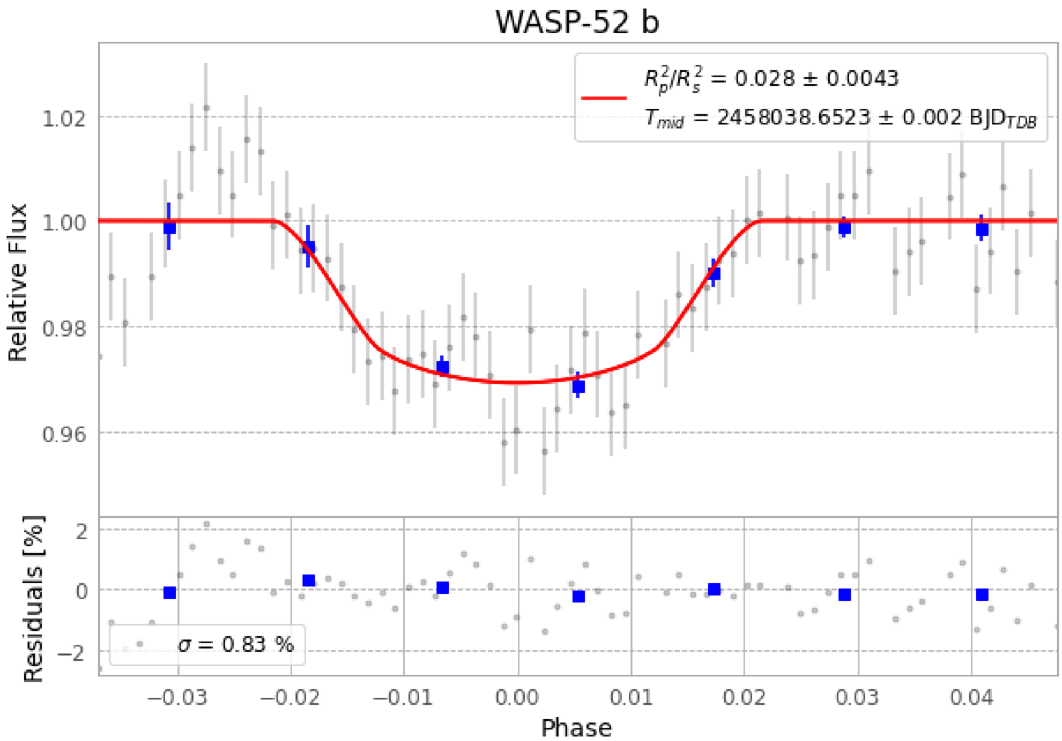

Figure 1 — FinalLightCurve_WASP-52 b_2017-10-11.png (SHA-256 9f7fdfa0cdcc6087…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [325, 298], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5e58bd3fe3162b46…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

72 FITS frames (47 MB), `WASP-52171012015713.FITS` → `WASP-52171012053012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171012.log (85d0d3e23c0c…), FinalLightCurve_WASP-52 b_2017-10-11.png (9f7fdfa0cdcc…), FinalLightCurve_WASP-52 b_2017-10-11.csv (e0c2f1b599dc…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)